

Oxidative Stress and Antioxidant Defence

Antioxidant enzymes (SOD, catalase, glutathione peroxidases), synthesis of glutathione (GCLC, GCLM), NRF2 and FOXO3, phase detoxification II (NQO1, GST), paraoxonase, heme-oxygenase and sources of oxidants (NOS3, MPO, CYBA)

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

Contents

Section	Page
1. Summary	3
2. How to read this report	4
3. Polygenic scores	5
4. Variants by theme	6
5. Favourable	9
6. Unfavourable	10
7. Mechanisms of ageing	11
8. Practical suggestions	12
9. Limitations	13
10. Methods and sources	14
11. Annex: all variants analysed	15

1. Summary

What was seen. 31 SNPs were read in 19 genes (23 chip, 7 imputed, 1 unread) and 5 polygenic scores. The portrait is of master regulators and strong main enzymes, with a phase detoxification II and a weaker paraoxonase.

Strong points. Homozygous FOXO3 for the longevity allele in the main 4 SNPs (and 1 favourable allele in rs12212067); **NRF2 –617C/C** (without the allele that stops self-induction); **SOD2 Ala/Ala** (MnSOD more active); **GCLC –129C/C** (glutathione synthesis without the weak allele); **NQO1 *1/*1** (full activity); GSTP1 without Val114; **HMOX1 –413A/A**; APOE without ε4. The longevity score (score A) is in p92,4.

Weak points. Probable null GSTM1 (A/A deletion marker) and **GSTA1 *B/*B** (~3-4x minus GSTA1 in the liver): the conjugation of epoxides and lipid peroxidation products depends more on GSTP1 and glutathione. PON1 with low expression (–108T/T and Met55/Met55), partially compensated by Gln192Gln. **GPX1 Pro198Leu** and **GCLM –588T** heterozygous; **NRF2 –653G** (1 copy) and –651A (called the uncertain chip). **MPO –463G/G** (more myeloperoxidase) and NOS3 –786C and Asp298 heterozygous (less nitric oxide).

Reading. The SOD2 Ala/Ala forms H₂O₂ faster and the GPX1 Leu198 removes it a little slower: the equilibrium depends on catalase (typical), glutathione and selenium. Null GSTM1 and GSTA1 *B/*B make exposure to smoke, burned meat and pollutants more relevant than to the average.

2. How to read this report

- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (Beagle 5.5, public panel of genomes complete to 30x) with quality $DR2 \geq 0.8$. Under this threshold, or outside the imputed zone, SNP is treated as SNP not found in the data.
- **Variants:** 31 SNPs: 23 of the chip, 7 imputed and 1 not found. All genotypes were reread directly in the raw DNA and in the imputation for this report.
- **Strand:** All genotypes are in the GRCh38 strand +. 0 SNPs needed conversion to complementary strand and 2 are palindromic (A/T or C/G). When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Chromosome X:** Sérgio F has only one copy of X outside the pseudoautosomal regions, so the SNPs of X appear with a single letter (hemizotic).
- **Polygenic scores (PGS):** 5 PGS Catalog scores calculated locally with 120.035 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$), calculated with exactly the same variants. Percentage ≥ 80 : MAIS PROVABLE/ALTO; ≤ 20 : MENOS PROVABLE/BAIXO; between 20 and 80: TIPIC.
- **Judgment:** Each result is divided into Favourable or Unfavourable (or typical/without judgment). A genotype is a probability, not a destination; almost all effects are small and add to the lifestyle.
- **Evidence:** Strong : replicated and with clear clinical effect; Moderate = GWAS/consistent meta-analysis, modest effect; Weakness : small or mechanistic studies.

3. Polygenic scores

Percentages in the reference population. The bilirubin is an endogenous antioxidant (Gilbert's syndrome, UGT1A1*28 heterozygous, is already known). The longevity score B has only 4 variants.

Endogenous antioxidants and longevity

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Total bilirubin (score A)	PGS002160 (Am J Hum Genet 2021)	118.548 120.068 (weighted coverage 100%)	69,5	TIPIC	typical
Total bilirubin (score B)	PGS000697 (Nat Genet 2021)	1.061 1.159 (weighted coverage 96%)	71,4	TIPIC	typical
Uric acid	PGS000126 (Nat Genet 2019)	110 114 (weighted coverage 91%)	40,9	TIPIC	without judgment
Longevity (score A)	PGS000906 (J Gerontol A 2021)	312 330 (weighted coverage 90%)	92,4	MAIS PROVABLE	favourable
Longevity (score B, 4 variants)	PGS002795 (Clin Infect Dis 2021)	4 4 (weighted coverage 100%)	47,2	TIPIC	typical

4. Variants by theme

For each SNP: genotype (strand +), origin, effect allele and number of copies, judgment and evidence. In the "protective" SNPs, having the allele is favourable.

Primary antioxidant enzymes

SOD2 (mitochondria) and SOD3 (out of cells) transform superoxide into H₂O₂; catalase and glutathione peroxidases (GPX1, GPX4) transform H₂O₂ into water. The balance between the two steps is what counts.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
SOD2 (MnSOD)	rs4880	Val16Ala (Ala = G in strand +)	G/G	chip	G (2 copy/s)	favourable	Ala16: more efficient mitochondrial import and ~30–40% more active MnSOD (the H ₂ O ₂ formed must be removed by GPX1/ catalase)	High
SOD3 (EC-SOD)	rs1799895	Arg213Gly	SNP not found in the data	SNP not found in the data (DR2<0.8 imputation)	G faux copy/s)	without data	Gly213 releases EC-SOD from the vascular wall; associated with coronary disease	High
CAT (catalase)	rs1001179	–262C>T	C/C	chip	T (0 copy/s)	neutral	Promoter; discordant direction between studies	Moderate
GPX1	rs1050450	Pro198Leu	G/A	chip	A (1 copy/s)	unfavourable (light)	Leu198: ~20–40% less activity and lower selenium response	High
GPX4	rs713041	c.718T>C (SECIS)	T/C	chip	T (1 copy/s)	without judgment	Changes the selenium incorporation priority; heterozygous = intermediate	Moderate

Glutathione: synthesis and recycling

Glutathione is the main antioxidant within cells. GCL (subunits GCLC and GCLM) is the limiting step of its synthesis; NAC provides cysteine.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
GCLC	rs17883901	–129C>T	G/G	chip	A (0 copy/s)	favourable	–129T reduces 40–50% the limiting step of glutathione synthesis	High
GCLM	rs41303970	–588C>T	G/A	chip	A (1 copy/s)	unfavourable (light)	–588T reduces ~40% the promoter of the modulatory subunit: less glutathione	High

Master regulators (NRF2 and FOXO3)

NRF2 binds more than 200 antioxidant and detoxification genes; FOXO3 induces SOD2, catalase and autophagy and is the most replicated longevity gene.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
NFE2L2 (NRF2)	rs6721961	–617C>A	G/G	imputation DR2=1.00	T (0 copy/s)	favourable	–617A reduces the self-induction of NRF2 (the strongest effect of the promoter)	High
NFE2L2 (NRF2)	rs35652124	–653A>G	T/C	chip	C (1 copy/s)	unfavourable (light)	–653G modestly reduces the promoter	Moderate

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
NFE2L2 (NRF2)	rs6706649	-651G>A	T/T	chip	T (2 copy/s)	unfavourable	-651A reduces the promoter (called chip with uncertain quality)	Low
FOXO3	rs2802292	intronic	G/G	chip	G (2 copy/s)	favourable	G = more replicated longevity allele; FOXO3 induces SOD2, catalase and autophagy	High
FOXO3	rs2764264	intronic	C/C	chip	C (2 copy/s)	favourable	C = longevity allele	High
FOXO3	rs13217795	intronic	C/C	chip	C (2 copy/s)	favourable	C = longevity allele	High
FOXO3	rs2802288	intronic	A/A	chip	A (2 copy/s)	favourable	A = longevity allele	High
FOXO3	rs12212067	intronic	T/G	chip	G (1 copy/s)	favourable	G: more TGF-β1 and less TNF in monocytes (lighter inflammatory course)	Moderate

II phase detoxification (NQO1 and GST)

NQO1 reduces quinones without radical formation; glutathione S-transferases (GSTP1, GSTA1, GSTM1) combine electrolytes and lipid peroxidation products.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
NQO1	rs1800566	Pro187Ser (*2)	G/G	chip	A (0 copy/s)	favourable	*2: almost zero enzyme (reduces quinones and keeps ubiquinol and vitamin E reduced)	High
NQO1	rs1131341	Arg139Trp (*3)	G/G	chip	A (0 copy/s)	favourable	*3: splicing altered, less activity	High
GSTP1	rs1695	Ile105Val	A/G	chip	G (1 copy/s)	without judgment	Val105: substrate-dependent activity	High
GSTP1	rs1138272	Ala114Val	C/C	chip	T (0 copy/s)	favourable	Val114: less activity	High
GSTA1	rs3957357	-69 (*B)	A/A	chip	A (2 copy/s)	unfavourable	B/B: ~3-4x minus GSTA1 in liver (GST with peroxidase activity)	High
GSTM1	rs366631	deletion marker	A/A	chip	A (2 copy/s)	unfavourable	Null GSTM1 marker (without enzyme GSTM1: epoxides, PAH and peroxidation products); does not measure the number of copies	Moderate

Paraoxonase and hemeoxygenase

PON1, bound to HDL, protects lipoproteins from oxidation; heme-oxygenase (HMOX1) forms bilirubin and CO, endogenous antioxidants.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
PON1	rs662	Gln192Arg	T/T	chip	C (0 copy/s)	mixed	QQ (without Arg): better protects lipoproteins from peroxidation, but less organophosphate hydrolyses	High
PON1	rs854560	Leu55Met	T/T	imputation DR2=1.00	T (2 copy/s)	unfavourable	Met55: less PON1 in serum and less stable	High
PON1	rs705379	-108C>T	A/A	imputation DR2=0.97	A (2 copy/s)	unfavourable	-108T: ~2x less expression (main genetic determinant of PON1 levels)	Moderate

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
HMOX1	rs2071746	–413A>T	A/A	imputation DR2=0.94	T (0 copy/s)	favourable	–413A: plus basal HO-1 (bilirubin and CO antioxidants); T gives less	Moderate-low

Oxidant sources and nitric oxide

Oxidant sources (neutrophil myeloperoxidase, NADPH-oxidase) and eNOS, which form nitric oxide; when decoupled, it forms superoxide. APOE ε4 as reference for cerebral lipid peroxidation.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
NOS3 (eNOS)	rs2070744	–786T>C	C/T	chip	C (1 copy/s)	unfavourable (light)	–786C: ~50% less activity of eNOS promoter and less nitric oxide	High
NOS3 (eNOS)	rs1799983	Glu298Asp	T/G	imputation DR2=1.00	T (1 copy/s)	unfavourable (light)	Asp298: eNOS more sensitive to cleavage and decoupling	Moderate
NOS3 (eNOS)	rs3918226	–665C>T	C/C	chip	T (0 copy/s)	favourable	Allele of promoter associated with endothelial dysfunction in GWAS	High
MPO	rs2333227	–463G>A (G = C in strand +)	C/C	imputation DR2=1.00	C (2 copy/s)	unfavourable	–463G: more myeloperoxidase (more hypochlorous acid and oxidation of LDL)	High
CYBA (p22phox)	rs4673	His72Tyr (242C>T)	G/G	chip	A (0 copy/s)	neutral	242T: less NADPH-oxidase activity in part of the studies	Low
APOE	rs429358	ε4 (Cys112Arg)	T/T	imputation DR2=0.99	C (0 copy/s)	favourable	ε4: lower antioxidant capacity and more cerebral lipid peroxidation	High

5. Favourable

- **FOXO3**: homozygous for the longevity allele in rs2802292 (G/G), rs2764264 (C/C), rs13217795 (C/C) and rs2802288 (A/A); 1 favourable allele in rs12212067.
- **NRF2 –617C/C** (rs6721961 G/G, imputation DR2 1.00): without the allele that reduces self-induction.
- **SOD2 Ala16/Ala16** (rs4880 G/G): MnSOD more active.
- **GCLC –129C/C**: glutathione synthesis without the weak allele.
- **NQO1 *1/*1** (without *2 or *3) and GSTP1 without Val114.
- **HMOX1 –413A/A** (imputed): more basal heme-oxygenase.
- **NOS3 –665** without the risk allele; APOE without ε4; **catalase –262C/C** (typical).
- **Longevity (score A) p92,4**; bilirubin p69,5–71.4 (typical, to tend to high).

6. Unfavourable

- **Probable null GSTM1** (rs366631 A/A; marker, does not measure copies) and **GSTA1 *B/*B** (rs3957357 A/A).
- **Low expression PON1:** –108T/T (rs705379 A/A) and Met55/Met55 (rs854560 T/T), both imputed.
- **GPX1 Pro198Leu** (1 copy): less glutathione peroxidase and less selenium response.
- **GCLM –588T** (1 copy): less subunit modulating GCL.
- **NRF2:** –653G (1 copy) and –651A (T/T; chip call with uncertain quality).
- **MPO –463G/G** (imputed): more myeloperoxidase.
- **NOS3:** –786C (1) and Asp298 (1), less endothelial nitric oxide.

7. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

Mechanism	Genetic finding	Meaning
Oxidative stress	SOD2 Ala/Ala, GCLC –129C/C, NQO1 1/1 and NRF2 –617C/C strong; Null GSTM1, GSTA1 B/B, GPX1 Leu198 and lower PON1	mixed
Redox imbalance	Rapid SOD2 + GPX1 Leu198: more transient H2O2; glutathione and selenium decide the balance	unfavourable (light)
Mitochondrial dysfunction	MnSOD more active (SOD2 Ala/Ala); NQO1 keeps the ubiquinol reduced	favourable
Autophagy / Deregulated Macroautophagy	Homozygous FOXO3 of longevity (induces autophagy)	favourable
Nutrient Sensing Deregulation	FOXO3 (via insulin/IGF-1) with longevity alleles; longevity score p92,4	favourable
Vascular Aging	NOS3 –786C and Asp298 heterozygous; MPO –463G/G; Low expression PON1	unfavourable (light)
Arterial stiffness	Less endothelial nitric oxide (NOS3) and more oxidized LDL (MPO, PON1)	unfavourable (light)
DNA Damage and DNA Repair Deficit	GSTM1 null and GSTA1 B/B: less conjugation of epoxides and PAH (DNA fertilizers)	unfavourable
Chronic Inflammation	FOXO3 rs12212067 G (less TNF in monocytes); MPO –463G/G (more neutrophil oxidants)	mixed
Neuroinflammation	APOE without ε4 (less cerebral lipid peroxidation)	favourable

8. Practical suggestions

All suggestions are suggestion (not applied): the real stack of Sérgio F remains as it is, and any change should be discussed with the doctor.

1. **Avoid smoking and burned meat** (carbonized grilles, smoked, tobacco smoke, combustion at home): GSTM1 null and GSTA1 *B/*B conjugate worse PAH and epoxides. Cooking at the lowest temperature (stuttering, steam, oven). *Suggestion (not applied).*
2. **Crucifers almost every day** (broccoli, kale, rocket, cress; broccoli shoots): the sulforaphane activates NRF2 (–617C/C responds well) and induces the remaining GST. *Suggestion (not applied).*
3. **NAC** (already in the stack) and protein with cysteine: support glutathione (GCLM –588T, GPX1 Leu198). *Suggestion (not applied).*
4. **Selenium by feeding** (1–2 chestnuts per day or fish and eggs): cofactor of GPX1 and GPX4; do not add selenium supplements without measuring. *Suggestion (not applied).*
5. **Omega-3 3.2–4 g/day of EPA+DHA**, with olive oil and polyphenols from the feed: protect HDL and PON1 of low expression. *Suggestion (not applied).*
6. **Avoid organophosphates** (domestic insecticides, well-washed fruits and vegetables): Low expression PON1. *Suggestion (not applied).*
7. **Regular exercise** (active FOXO3, NRF2, SOD2 and eNOS) and vegetal nitrates (beet, rocket, spinach) for nitric oxide (NOS3 –786C). *Suggestion (not applied).*
8. **Antioxidants in megadoses** (vitamin E, vitamin C at high doses) are not recommended: with these genes, the strategy is to activate endogenous defenses, not replace them. *Suggestion (not applied).*
9. **Zero alcohol** (It is already the rule). *Suggestion (not applied).*

9. Limitations

- Small and added effects; do not measure real oxidative stress.
- **GSTM1**: The rs366631 marker suggests deletion, but does not measure the number of copies. **GSTT1** is not testable on the chip.
- **Unread**: SOD3 Arg213Gly (rs1799895; out of the chip and without imputation with $DR2 \geq 0.8$; a previous commercial reading gave C/C, without the rare allele) and functional repetitions (NOS3 VNTR, HMOX1 (GT)_n, GCLC GAG, GPX1 alanines).
- The NRF2 –651 (rs6706649) has called the doubtful chip.
- Seven SNPs come from imputation ($DR2$ 0.94–1.00).
- Conclusions only from genes; do not include blood analysis.

10. Methods and sources

- **Genes and SNPs:** 31 SNPs chosen from the previous analysis of oxidative stress (64 lines), maintaining functional variants; MTHFR and COMT are in the report Methylation, Homocysteine and Hill.
- **Genotypes:** Sérgio F chip (GRCh38, strand +), reread for this report. Absent SNPs come from local imputation (Beagle 5.5, public panel of complete 30x genomes), used only with $DR2 \geq 0.8$. No genetic data came out of the analysis computer.
- **Guidance:** Alleles of effect on strand +; in the genes of strand – (SOD2, MPO, PON1, CYBA, NQO1) the nomenclature of the gene was converted. 2 palindromic SNPs assumed in the strand +.
- **Polygenic scores:** PGS Catalog, calculated with the imputed genome; percentiles in the reference population.
- **Comparison with previous analysis (02-10-2026):** The 23 chip genotypes are equal; the 7 imputed (NRF2 –617, PON1 ×2, HMOX1, NOS3 Asp298, MPO, APOE) confirm the previous inference by close neighbors; the SOD3 passed to "unread".
- **Context Literature:** Willcox 2008 and Flachsbart 2009 (FOXO3), Marzec 2007 (NRF2), Bastaki 2006 (SOD2/GPX1), Huang 2009 (GSTM1 marker).

11. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Primary antioxidant enzymes	SOD2 (MnSOD)	rs4880	G/G	chip	G	2	strand +	Ala/Ala: MnSOD more active (depends on GPX1 and catalase)
Primary antioxidant enzymes	SOD3 (EC-SOD)	rs1799895	SNP not found in the data	SNP not found in the data (DR2<0.8 imputation)	G	—	—	SNP not found in the data
Primary antioxidant enzymes	CAT (catalase)	rs1001179	C/C	chip	T	0	strand +	C/C: common genotype, typical activity
Primary antioxidant enzymes	GPX1	rs1050450	G/A	chip	A	1	strand +	1 copy of risk allele
Primary antioxidant enzymes	GPX4	rs713041	T/C	chip	T	1	strand +	1 copy of effect allele
Glutathione: synthesis and recycling	GCLC	rs17883901	G/G	chip	A	0	strand +	without the risk allele
Glutathione: synthesis and recycling	GCLM	rs41303970	G/A	chip	A	1	strand +	1 copy of risk allele
Master regulators (NRF2 and FOXO3)	NFE2L2 (NRF2)	rs6721961	G/G	imputation DR2=1.00	T	0	strand +	without the risk allele
Master regulators (NRF2 and FOXO3)	NFE2L2 (NRF2)	rs35652124	T/C	chip	C	1	strand +	1 copy of risk allele
Master regulators (NRF2 and FOXO3)	NFE2L2 (NRF2)	rs6706649	T/T	chip	T	2	strand +	2 risk allele copies
Master regulators (NRF2 and FOXO3)	FOXO3	rs2802292	G/G	chip	G	2	strand +	2 copies of the protective allele
Master regulators (NRF2 and FOXO3)	FOXO3	rs2764264	C/C	chip	C	2	strand +	2 copies of the protective allele
Master regulators (NRF2 and FOXO3)	FOXO3	rs13217795	C/C	chip	C	2	strand +	2 copies of the protective allele
Master regulators (NRF2 and FOXO3)	FOXO3	rs2802288	A/A	chip	A	2	strand +	2 copies of the protective allele
Master regulators (NRF2 and FOXO3)	FOXO3	rs12212067	T/G	chip	G	1	strand +	1 copy of the protective allele
II phase detoxification (NQO1 and GST)	NQO1	rs1800566	G/G	chip	A	0	strand +	without the risk allele
II phase detoxification (NQO1 and GST)	NQO1	rs1131341	G/G	chip	A	0	strand +	without the risk allele
II phase detoxification (NQO1 and GST)	GSTP1	rs1695	A/G	chip	G	1	strand +	1 copy of effect allele

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
II phase detoxification (NQO1 and GST)	GSTP1	rs1138272	C/C	chip	T	0	strand +	without the risk allele
II phase detoxification (NQO1 and GST)	GSTA1	rs3957357	A/A	chip	A	2	strand +	2 risk allele copies
II phase detoxification (NQO1 and GST)	GSTM1	rs366631	A/A	chip	A	2	strand +	Probable null GSTM1 (marker)
Paraoxonase and hemeoxygenase	PON1	rs662	T/T	chip	C	0	strand +	Gln/Gln: better protection of lipoproteins; less defense against organophosphates
Paraoxonase and hemeoxygenase	PON1	rs854560	T/T	imputation DR2=1.00	T	2	strand + (palindromic; assumed +)	2 risk allele copies
Paraoxonase and hemeoxygenase	PON1	rs705379	A/A	imputation DR2=0.97	A	2	strand +	2 risk allele copies
Paraoxonase and hemeoxygenase	HMOX1	rs2071746	A/A	imputation DR2=0.94	T	0	strand + (palindromic; assumed +)	without the risk allele
Oxidant sources and nitric oxide	NOS3 (eNOS)	rs2070744	C/T	chip	C	1	strand +	1 copy of risk allele
Oxidant sources and nitric oxide	NOS3 (eNOS)	rs1799983	T/G	imputation DR2=1.00	T	1	strand +	1 copy of risk allele
Oxidant sources and nitric oxide	NOS3 (eNOS)	rs3918226	C/C	chip	T	0	strand +	without the risk allele
Oxidant sources and nitric oxide	MPO	rs2333227	C/C	imputation DR2=1.00	C	2	strand +	2 risk allele copies
Oxidant sources and nitric oxide	CYBA (p22phox)	rs4673	G/G	chip	A	0	strand +	without the protective allele (common genotype)
Oxidant sources and nitric oxide	APOE	rs429358	T/T	imputation DR2=0.99	C	0	strand +	without the risk allele